II B.E. (Computer Engineering)
Semester Examination December 2023
3CERG1: Computer Organization and Architecture

2024024

Time: 3Hrs. **Max. Marks : 60**

Note: Attempt all questions. Write answer of any two parts from each question. All Questions carry equal marks.

Q.1 (a) Differentiate between computer architecture and organization. Give two each of architectural and organizational features of a computer. What are the major components of a core in a multicore computer? **(6)**

(b) A benchmark program is executed on 400Mhz processor with following instruction mix **(6)**

Instruction Category	Instruction Count	Cycles per instruction
Arithmetic and logic	45,000	1
Load/store with cache hit	32,000	2
Branch	15,000	4
Memory reference with cache miss	8000	8

Calculate average CPI, average Execution time and MIPS for this program.Also Which instruction category takes longest and shortest execution time?

(c) (i) Write decimal equivalent of binary number A =11101011, binary number A is represented in two's complement form. **(6)**

(ii) Represent a twos complement 8 bit binary number A= 11101001 as 16 bit twos complement binary number

(iii) Represent floating point number 1101.111 in IEEE 754 format.

(d) Calculate P = A X B using Booths algorithm. Where A= 25 and B= -19. Assume numbers are represented in 6 bits binary two's complement representation.. **(6)**

Q.2 (a) Compare the characteristics of (i) SRAM, DRAM, SDRAM &DDRAM and (ii) PROM, EPROM and flash memory. **(6)**

(b) A ROM memory is organized into 2Mx8 chip

(i) Show the number of address lines and data lines in the chip.

(ii) Show the size of memory in bytes and bits.

(iii) Show the total number of chips required, if the memory size of 16Mx16 to be organized using this chip.

(c) Consider the main memory size as 16MB, cache size as 32KB and word size as 4byte. **(6)**

(i) Show the length of tag, line and word fields, in direct mapped cache Organization.

(ii) Show the length of tag and word fields in fully associative cache organization.

(iii) Show the length of tag, set and word fields in 4-way set-associative cache Organization.

(d) (i) Explain the placement of L1, L2 and L3 cache memory in a microprocessor chip. **(6)**

(ii) Describe write back and write through policy in the cache organization.

Q.3 (a) Compare the merits and demerits of following addressing modes. How is effective address calculated in these addressing modes? **(6)**

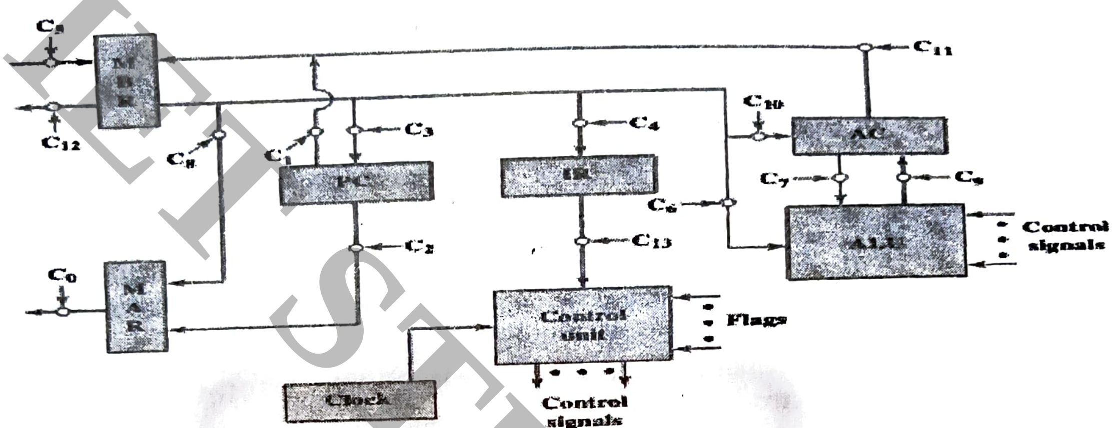
(i) Direct addressing
(ii) Register indirect
(iii) Relative Addressing
(iv) Base register addressing
(v) Indexing

(b) Given the following memory values and a one-address machine with an accumulator, what values do the following instructions load into the accumulator? Assume Word 20 contains 40, Word 30 contains 50, Word 40 contains 60 and Word 50 contains 70. **(6)**

a. LOAD IMMEDIATE 20
b. LOAD DIRECT 20
c. LOAD INDIRET 20
d. LOAD IMMEDIATE 30
e. LOAD DIRECT 30
f. LOAD INDIRECT 30

(c) Explain instruction execution process. Show all tasks of instruction execution as state machine **(6)**

(d) Following is the data paths and control signals generated by control unit for various components of CPU. **(6)**



i) Explain the role of various control signals (any three) in this diagram.
ii) What control signals will be active for following micro-operations

MAR ← (PC)
IR(Address) ← (MBR(Address))
MBR ← Memory
IR ← (MBR)

Q.4 (a) Explain the internal structure of an I/O module with the help of a block diagram. **(6)**

(b) When a device interrupt occurs, show the steps followed to handle the interrupt, also show how the content of program counter and stake pointer changes at the time of interrupt occur and return from interrupt to main program. **(6)**

(c) How flag register is different from a general purpose register? Which flag bit(s) will be affected if the last operation performed with 4 bit binary word is addition of +5 and +6. **(6)**

(d) What is the advantage of DMA? Show all the steps in DMA operation. **(6)**

Q.5 (a) How instruction pipelining reduces overall execution time. Explain by giving example. **(6)**

(b) What is pipeline data hazard? Explain three different types of data hazards. **(6)**

(c) How is microprogrammed control unit functions? Describe various functional blocks of microprogrammed control unit **(6)**

(d) Describe the Flynn's taxonomy of computer systems design. Differentiate SMP and NUMA architecture. **(6)**

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